



First Semester 2026-27

Course Handout Part II

Date: 1st August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number : MEL G624

Course Title : Advanced VLSI Architectures

Instructor-in-charge : Dr. Govind Prasad

Team of Instructors : Prof. Chandra Shekhar, Dr. Govind Prasad

- 1. Course description:** The course explores advanced VLSI processor architectures, their implementation strategies (microarchitectures) and associated memory hierarchies and their microarchitectures for high-performance system design. It also covers the various techniques utilized to leverage parallelism at hardware, application and compiler levels. Students will gain an understanding of how (beyond instruction-level parallelism) data-level parallelism, thread-level parallelism and task-level parallelism can be exploited to architect and implement different processing cores e.g.: Vector Processors, SIMD processors and Multimedia processors, Graphical Processing Units (GPU) and Multi-core processors, Domain Specific Processors, and Memory hierarchy organizations for the above processors. The course also explores the emerging research directions in VLSI processor design and system architectures.
- 2. Scope and Objective:** The objective of the course is to understand and gain an in-depth view of Advanced VLSI Architectures, specifically to study various processor architectures, associated microarchitectures, and memory organizations that have evolved to exploit different types of parallelism to increase processor performance. The scope of the course covers the following in detail:
 - a. Study of advanced processor architectures and associated implementation techniques (microarchitectures) and memory organization techniques
 - b. This includes the study of Vector, SIMD, GPU, and Domain Specific architectures and implementations
 - c. Study of architectures and implementations that exploit thread-level parallelism and task-level parallelism via multi-processing (includes multicore processors)
 - d. Study of memory hierarchy design for CPUs, vector processors, GPUs, and application domain-specific processors
- 3. Course learning outcomes:**
 - a. Explain the advantages of CORDIC in hardware implementation, especially in terms of multiplier-less architecture, low power, and area efficiency.

- b. **Analyze** the design principles of advanced VLSI processor architectures and their microarchitectural implementation strategies for high-performance computing systems.
- c. **Evaluate** various memory hierarchy designs and optimizations, including their microarchitectural aspects, in the context of different processor types.
- d. **Apply** concepts of parallelism—such as instruction-level, data-level, thread-level, and task-level parallelism—to optimize system performance.
- e. **Design** architectural solutions using specialized processing units such as SIMD processors, vector processors, multimedia processors, GPUs, multi-core systems, and domain-specific accelerators.
- f. **Explore** contemporary research trends and emerging technologies in VLSI processor and memory system architecture to identify potential areas for innovation.

4. Textbooks and references:

TB1. [Computer Architecture: A Quantitative Approach, by J.L. Hennessy & D.A. Patterson, Morgan Kaufmann., 6th, 2019.](#)

RB1. Next-Gen Computer Architecture: Till The End of Silicon - Version 2.0, by Smruti Ranjan Sarangi, White Falcon Publishing, 2023.

RB2. Computers as Components: 5th Edition, by Marilyn Wolf, Elsevier, 2022.

RB3. Modern Processor Design: Fundamentals of Superscalar Processors, John Paul Shen & Mikko.H, Lipasti, Tata McGraw-Hill, 2011.

RB4. J. E. Volder, "The CORDIC Trigonometric Computing Technique," in IRE Transactions on Electronic Computers, vol. EC-8, no. 3, pp. 330-334, Sept. 1959, doi: 10.1109/TEC.1959.5222693.

RB5. Journals & Conference Proceedings

5. Method of conduct of the course: The course will be conducted through a combination of lectures, tutorials, seminars, and project-based learning to facilitate both theoretical understanding and practical skills development.

6. Course Plan: weekly schedule for 14 weeks with 3 lectures per week

Week no.	Module	Lec. no.	Ref.	Learning outcomes
1	CORDIC algorithm	01-03	RB3 & RB4	Compute trigonometric and other functions using only shifts and additions. Design efficient VLSI architectures for implementing CORDIC in real-time embedded and signal processing systems.
2	Fundamentals of Quantitative Design and Analysis	04-06	TB1-1.2 to 1.3 & TB1-A	Classification of computers from an architectural point of view, Quantitative principles of performance measurement.



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3	Memory Hierarchy Design	07-13	TB1-2.1 to 2.3 & TB1-B.1 to B.2, B.4	Understand the concept of memory hierarchy, Virtual memory, analyze different memory technologies, Design and optimization strategies.
4				
5	Pipeline architectures	14	TB1- C.1 to C.3	Understand and apply the principles of instruction pipelining, recognize and manage pipeline hazards, and evaluate their performance impact on modern RISC architectures.
6	Introduction to Parallel Processing, ILP	15-19	TB1-3.1 to 3.3 & 3.11	Understand the fundamentals of ILP, identify the main challenges in exploiting ILP, and apply basic compiler techniques to improve ILP
7				
8	Data-Level Parallelism in Vector, SIMD	20-25	TB1- 4.1 to 4.3	Vector architectures and implementations. Multimedia architectures and implementations.
9				
10	GPU Architectures	26-31	TB1- 4.4	Understand the architecture and functioning of modern GPUs
11				
12	TLP & multicores	32-35	TB1- 5.1 to 5.8	Understand the concepts and significance of TLP in modern computing systems. Gain knowledge of different multiprocessor architectures, including centralized shared memory and distributed shared memory systems
13				
14	Multiprocessor interconnects	36-37	Class Notes	Understand the fundamentals and significance of NoC architectures in modern SoC designs.
	Application / Domain Specific Processor design	38	TB1- 7.1 to 7.3 & Class Notes	Application / Domain Specific Processor design and implementation. Recent (and research) architectures and their implementations for machine learning.

7. Evaluation Scheme:

Evaluation Component	Duration	Marks (100) Weightage	Date & Time	Nature of Component
Mid-Semester Exam	90 Min	25M	06/10/2026	Closed Book
Take-home Assignments	NA	40M	To be announced	Open Book
Comprehensive Exam	180 Min	35M	05/12/2026	Closed Book

8. Chamber consultation hours: Friday, 4 PM to 6 PM (Room no: FD-II 2219-D)

9. Make-up Policy: Make-up for any component will be given only in genuine cases. In all cases, prior intimation must be given to the IC.

Instructor-in-charge

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